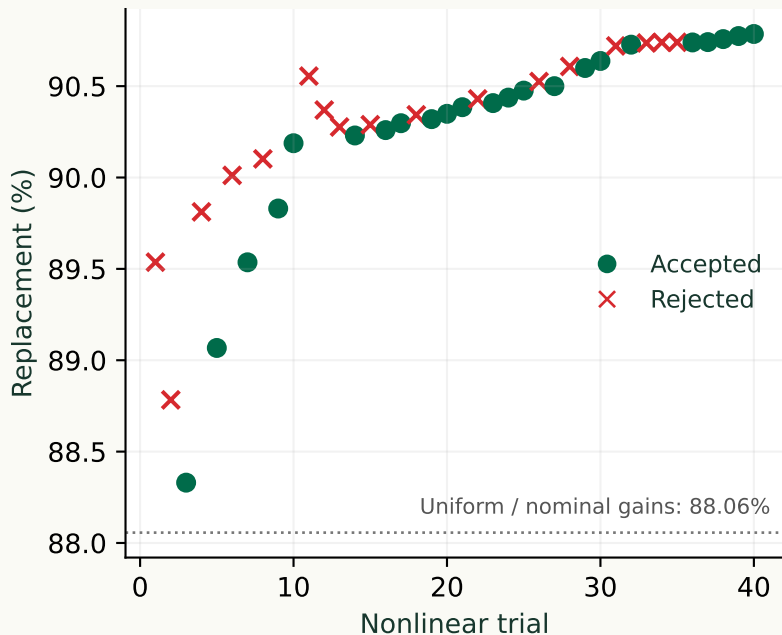


BEYOND NODAL DAMPING

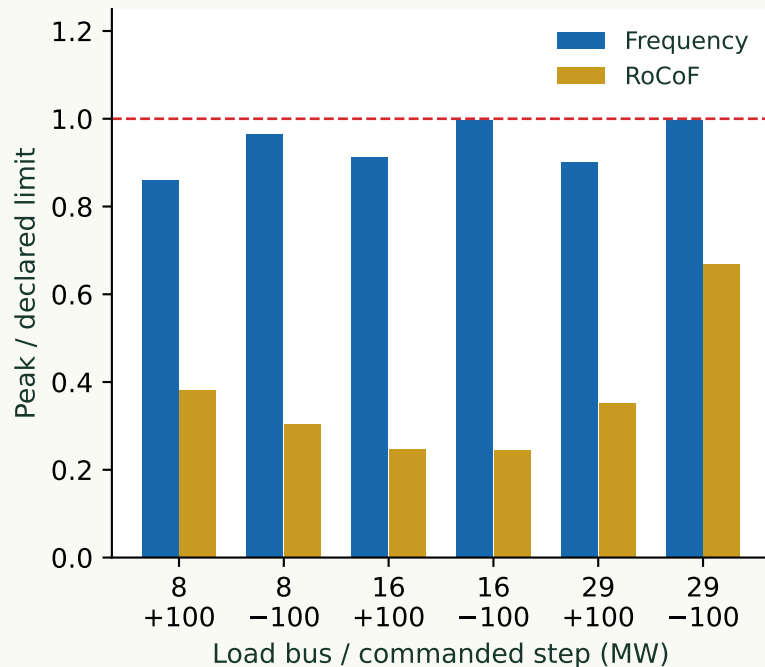
Nonlinear SG-to-GFL co-design — preliminary IEEE-39 instance

90.785% GFL replacement | 497.84 MW SG retained

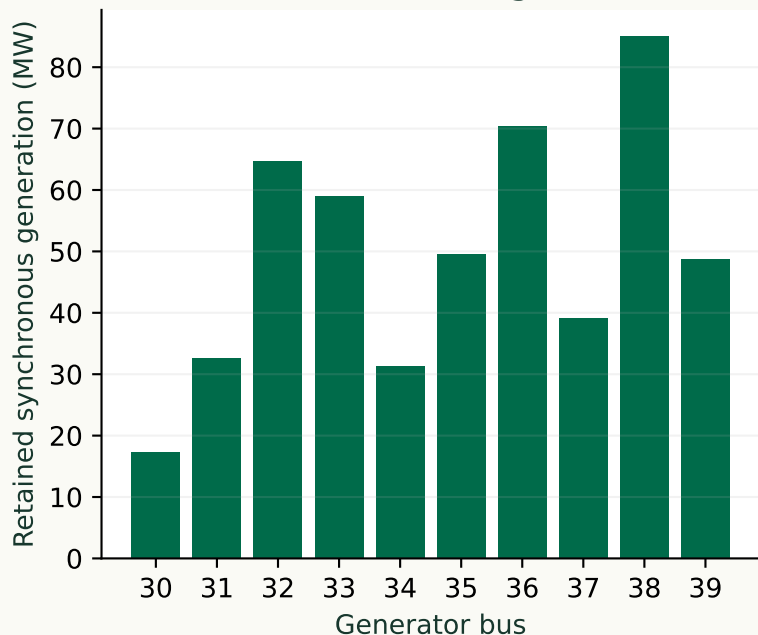
A Accept the trajectory, not just the poles



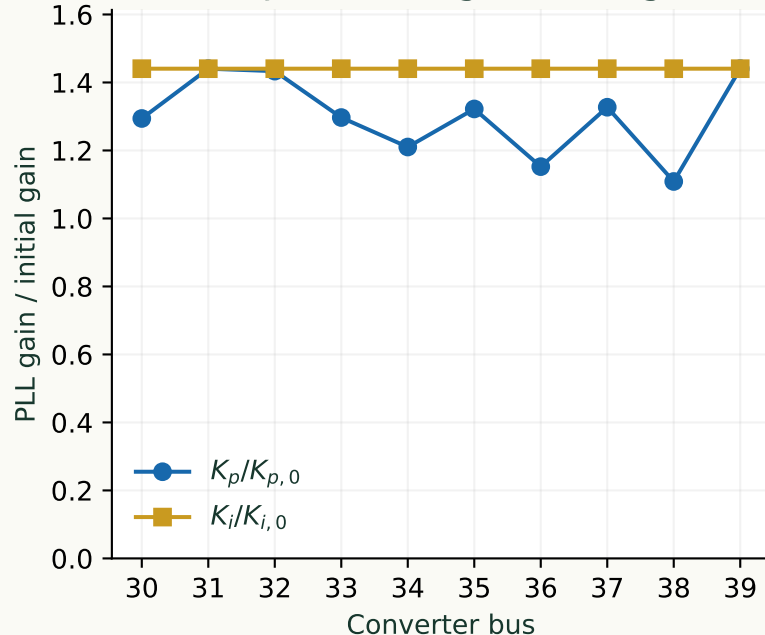
B Six nonlinear disturbance cases



C Location is a design variable



D Joint local gain tuning



Physical DC-supply convention • all 39 buses monitored • 60 s • 0.5 s measurement window

Best feasible found; global optimality and a nonlinear region-of-attraction certificate are not established.

Voltage band and actuator margin are declared experiment settings. Equipment current / energy limits are pending.